

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Write a program using JavaScript (arrays, functions)

2. AIM

To write and execute a JavaScript program using arrays and functions.

To implement functions for adding, displaying, and managing items in a shopping list.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic concepts of JavaScript arrays and functions.
 - To create and manipulate an array to store multiple shopping items.
 - To implement functions for adding and displaying items.
 - To use array methods such as `push()`, `join()`, and `length`.
 - To use conditional statements to display appropriate messages based on the number of items.
 - To use JavaScript functions to display the current date and continuously updated time.
 -
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4. THEORY

JavaScript is a programming language used to add logic, functionality, and interactivity to webpages. In this application, JavaScript is used to create an array for storing shopping items and functions for adding, displaying, and managing the items.

The program uses JavaScript concepts such as **arrays, functions, array methods, conditional statements, and the Date object**.

- **Arrays** are used to store multiple shopping items.
- **Functions** are used to perform specific tasks such as adding items and displaying the list.
- **push()** is used to add new items to the array.
- **join()** is used to display the array items as a string.
- **length** is used to find the total number of items.
- **Conditional statements** are used to display messages based on the number of items.
- **Date object** is used to display the current date and time.
- **setInterval()** is used to update the displayed time every second.
- **DOM methods** such as **document.getElementById()** are used to access and update HTML elements.

The basic flow of the program is:

User Input → JavaScript Function → Array Processing → Conditional Checking → DOM Update → Displayed Output

5. ALGORITHM / PROCEDURE

1. Create an HTML file and set up the structure with input fields for 'name' and 'marks', an 'Add Student' button, and output sections for the student list, statistics, and passed students.
2. Link the CSS file and reference the external JavaScript file.
3. Initialize an empty **students** array to store student objects.
4. Define the **addStudent()** function to capture user input, validate that fields are filled and marks are between 0-100, and add the new student object to the **students** array.
5. Call **displayStudents()**, **calculateStatistics()**, and **displayPassedStudents()** within **addStudent()** to update the interface.
6. Define **displayStudents()** to iterate through the array and update the 'studentList' element in the DOM.
7. Define **calculateStatistics()** to compute the total students, average marks, highest marks, and lowest marks using **map()**, **reduce()**, **Math.max()**, and **Math.min()** methods.
8. Define **displayPassedStudents()** to filter students with marks ≥ 40 and display them in the 'passedStudents' element.

9. Attach a 'click' event listener to the 'Add Student' button.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <title>Student Marks Manager</title>

    <link rel="stylesheet" href="style.css">

</head>

<body>

    <div class="container">

        <h1>Student Marks Manager</h1>

        <div class="input-section">

            <input
                type="text"
                id="nameInput"
                placeholder="Enter student name"
            >

            <input
```

```
        type="number"
        id="marksInput"
        placeholder="Enter marks"
        min="0"
        max="100"
    >

    <button id="addBtn">
        Add Student
    </button>

</div>

<h2>Student List</h2>

<div id="studentList"></div>

<div class="statistics">

    <h2>Statistics</h2>

    <p id="totalStudents">
        Total Students: 0
    </p>

    <p id="averageMarks">
        Average Marks: 0
    </p>

    <p id="highestMarks">
        Highest Marks: 0
    </p>

    <p id="lowestMarks">
        Lowest Marks: 0
    </p>

</div>
```

```
<div class="passed-section">

    <h2>Passed Students</h2>

    <div id="passedStudents">
        No students available.
    </div>

</div>

</div>

<script src="script.js"></script>

</body>

</html>
```

Style.css

```
body {
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
    margin: 0;
    padding: 40px;
}

.container {
    width: 700px;
    margin: auto;
    background-color: rgb(158, 248, 135);
    padding: 30px;
    border-radius: 10px;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.2);
}

h1 {
    text-align: center;
}
```

```
.input-section {
  display: flex;
  gap: 10px;
  margin-bottom: 30px;
}

input {
  flex: 1;
  padding: 12px;
  border: 1px solid #ccc;
  border-radius: 5px;
  font-size: 15px;
}

button {
  padding: 12px 20px;
  border: none;
  border-radius: 5px;
  background-color: #333;
  color: white;
  cursor: pointer;
}

button:hover {
  background-color: #555;
}

.student-card {
  display: flex;
  justify-content: space-between;
  padding: 12px;
  margin-bottom: 10px;
  border: 1px solid #ddd;
  border-radius: 5px;
}

.statistics,
.passed-section {
  margin-top: 30px;
  padding: 20px;
  border: 1px solid #ddd;
}
```

```
    border-radius: 8px;
    background-color: #fafafa;
}

.statistics p {
    font-size: 17px;
}

#passedStudents {
    font-size: 17px;
}
```

script.js

```
const students = [];

const nameInput = document.getElementById("nameInput");

const marksInput = document.getElementById("marksInput");

const addBtn = document.getElementById("addBtn");

const studentList = document.getElementById("studentList");

const totalStudents = document.getElementById("totalStudents");

const averageMarks = document.getElementById("averageMarks");

const highestMarks = document.getElementById("highestMarks");

const lowestMarks = document.getElementById("lowestMarks");

const passedStudents = document.getElementById("passedStudents");

function addStudent() {
```

```
const name = nameInput.value.trim();

const marks = Number(marksInput.value);

if (name === "" || marksInput.value === "") {

    alert("Please enter student name and marks.");

    return;
}

if (marks < 0 || marks > 100) {

    alert("Marks must be between 0 and 100.");

    return;
}

const student = {
    name: name,
    marks: marks
};

students.push(student);

nameInput.value = "";

marksInput.value = "";

displayStudents();

calculateStatistics();

displayPassedStudents();
}
```

```
function displayStudents() {

    studentList.innerHTML = "";

    if (students.length === 0) {

        studentList.textContent = "No students available.";

        return;
    }

    students.forEach(function(student, index) {

        const studentCard = document.createElement("div");

        studentCard.className = "student-card";

        const studentText = document.createElement("span");

        studentText.textContent =
            (index + 1) +
            ". " +
            student.name +
            " - " +
            student.marks;

        studentCard.appendChild(studentText);

        studentList.appendChild(studentCard);

    });
}
```

```
function calculateStatistics() {

    const total = students.length;

    totalStudents.textContent =
        "Total Students: " + total;

    if (total === 0) {

        averageMarks.textContent = "Average Marks: 0";

        highestMarks.textContent = "Highest Marks: 0";

        lowestMarks.textContent = "Lowest Marks: 0";

        return;
    }

    const marksArray = students.map(function(student) {

        return student.marks;

    });

    const totalMarks = marksArray.reduce(
        function(sum, marks) {

            return sum + marks;

        },
        0
    );

    const average = totalMarks / total;
```

```
const highest = Math.max(...marksArray);

const lowest = Math.min(...marksArray);

averageMarks.textContent =
  "Average Marks: " + average.toFixed(2);

highestMarks.textContent =
  "Highest Marks: " + highest;

lowestMarks.textContent =
  "Lowest Marks: " + lowest;
}

function displayPassedStudents() {

  const passed = students.filter(function(student) {

    return student.marks >= 40;

  });

  if (passed.length === 0) {

    passedStudents.textContent =
      "No students have passed.";

    return;

  }

  passedStudents.innerHTML = "";
```

```
passed.forEach(function(student) {

    const paragraph = document.createElement("p");

    paragraph.textContent =
        student.name +
        " - " +
        student.marks;

    passedStudents.appendChild(paragraph);

});

}

const findStudent = (name) => {

    return students.find(function(student) {

        return student.name.toLowerCase() ===
            name.toLowerCase();

    });

};

addBtn.addEventListener("click", addStudent);

displayStudents();

calculateStatistics();

displayPassedStudents();
```

7. CODE EXPLANATION

JavaScript / HTML Component	Description
const students = []	Creates an empty array to store student objects.
function addStudent()	Validates inputs and adds a new student object to the array.
Number()	Converts the marks input string to a number.
students.push(student)	Appends a new student object to the array.
displayStudents()	Iterates through the array and updates the student list UI.
calculateStatistics()	Calculates total students, average, highest, and lowest marks.
map()	Creates an array of only the marks from the student objects.
reduce()	Calculates the sum of all marks in the array.
Math.max() / Math.min()	Finds the highest and lowest values in the marks array.
filter()	Selects students who scored 40 or above.
displayPassedStudents()	Updates the UI with the list of passed students.
document.getElementById()	Accesses HTML elements by their ID.
innerHTML / textContent	Dynamically updates the content of the webpage elements.
alert()	Displays validation error messages.

8. TEST CASES AND EXECUTION DATA

Test Case	Test Scenario	Input / Action	Expected Result	Actual Result	Status
TC-01	Open webpage	Open HTML file in browser	Shopping List webpage should load	Webpage loaded correctly	Pass
TC-02	Add an item	Enter Milk and click Add Item	Item should be added successfully	successfully	Pass
TC-03	Add multiple items	Enter Bread , Rice , and Fruits	All items should be stored in the array	Items stored successfully	Pass
TC-04	Display shopping list	Click Show List	All entered items should be displayed	Items displayed correctly	Pass
TC-05	Check item count	Add 3 items and click Show List	Total Items should display 3	Total Items displayed as 3	Pass
TC-06	Empty input	Click Add Item without entering an item	Alert should ask the user to add an item	Alert displayed correctly	Pass
TC-07	Check list message	Add 5 or more items and click Show List	Message should indicate many items need to be bought	Correct message displayed	Pass
TC-08	Check date and time	Open the webpage	Current date and time should be displayed	Date and time displayed correctly	Pass
TC-09	Check live time	Keep the webpage open	Time should update every second	Time updated successfully	Pass

9. OUTPUT

Student Marks Manager

Add Student

Student List

No students available.

Statistics

Total Students: 0

Average Marks: 0

Highest Marks: 0

Lowest Marks: 0

Passed Students

No students have passed.

Student Marks Manager

Add Student

Student List

1. Bhuvan - 80

Statistics

Total Students: 1

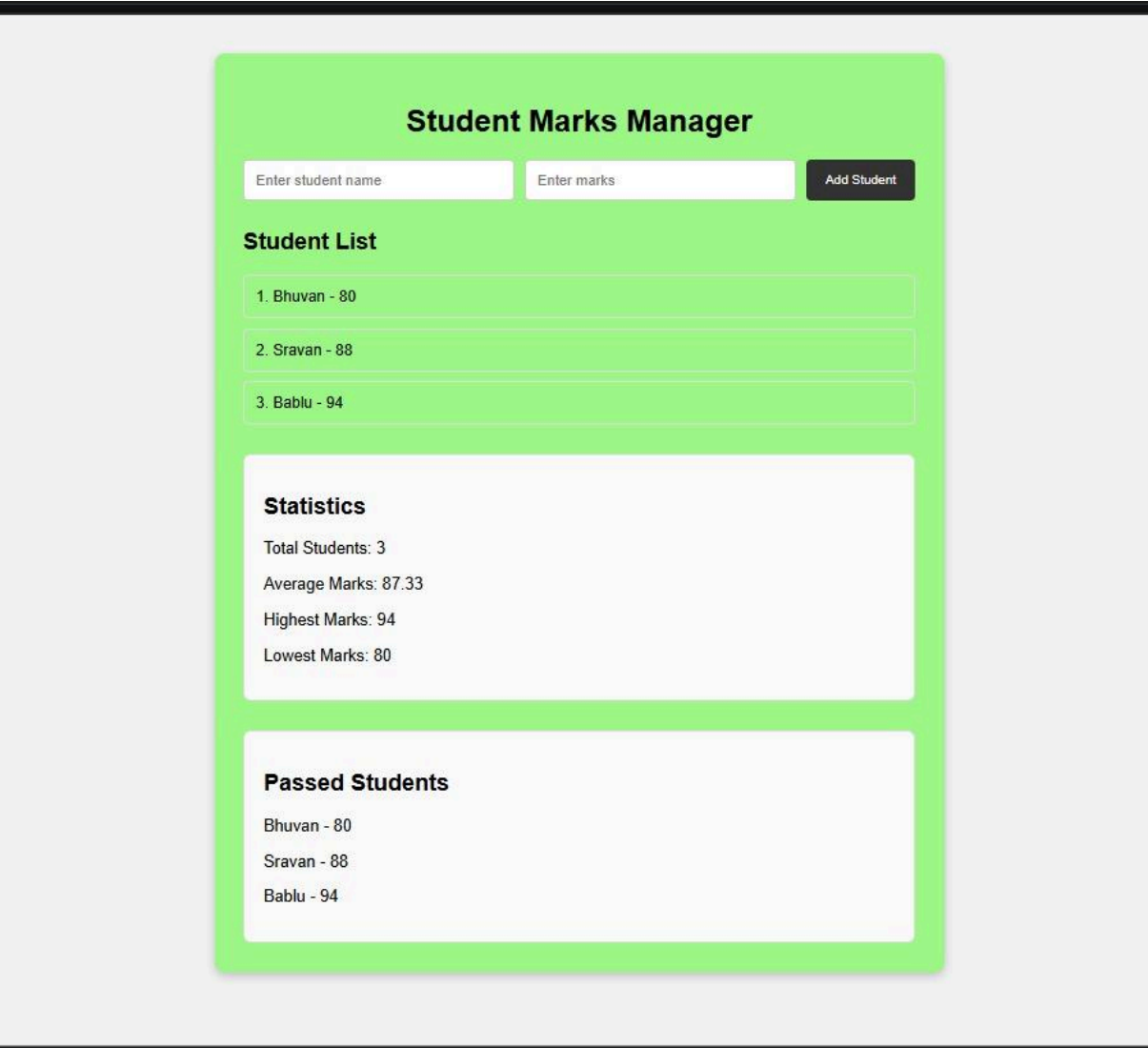
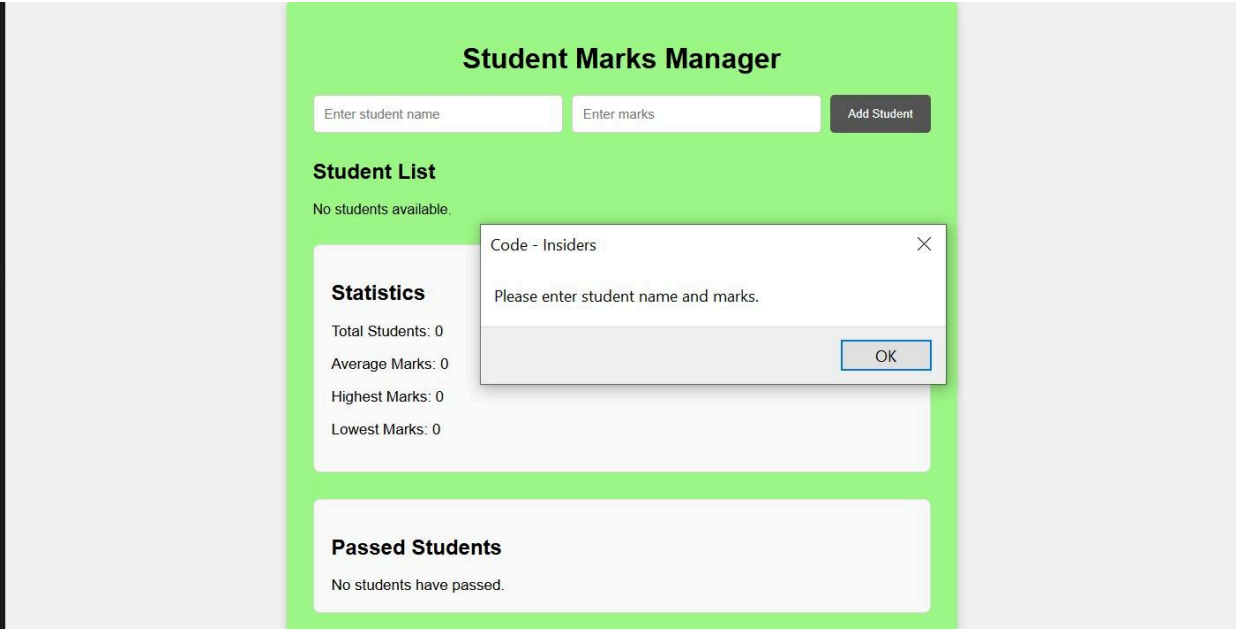
Average Marks: 80.00

Highest Marks: 80

Lowest Marks: 80

Passed Students

Bhuvan - 80



10. RESULT

Thus, a **JavaScript program using arrays and functions** was successfully developed. The program stores and displays shopping items using arrays, performs operations using functions, calculates the total number of items, and displays the current date and continuously updated time. The program was successfully tested using multiple test cases.